

ST4069

Multivariate Methods for Data Analysis

Note: Approval needed from Academic School/Department before you can take this module.

Subject Area: ST - Statistics

Credit Weighting: 10

The notional student workload for 5 credits is 125 hours

Semester(s)

Semester 2 (Jan-May)

No. of Students

Min 5, Max 25

Pre-requisite(s)

ST2053 - Introduction to Regression Analysis **AND**
ST2054 - Probability and Mathematical Statistics

Co-requisite(s)

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Teaching Methods

- 48 x 1 Hr(s) Lectures
- 12 x 1 Hr(s) Practicals/Laboratories
- 12 x 1 Hr(s) Research Training/Independent Enquiry

Module Co-ordinator

Prof Finbarr O'Sullivan - Department of Statistics

Lecturer(s)

- Staff - Department of Statistics

Guest Lecturer(s)

None

Module Goal

To develop basic tools for analysis of multivariate data.

Module Content

Representation of multivariate data with examples. Multidimensional random variables with emphasis on the Gaussian. Directional data. Dimension reduction. The Multivariate Linear Model. Model classification. Mixture Models and Clustering. Bayesian optimization.

Learning Outcomes

On successful completion of this module, students should be able to:

1. Describe and apply standard tools for summarising multivariate data;
2. Describe and apply algorithms for model-based and model-free clustering
3. Describe and apply methods of analysis for correlated outcome data with specific attention to repeated measures and longitudinal data.
4. Describe and apply discriminant analysis, principal components and factor analysis.
5. Demonstrate practical familiarity with relevant software tools (R, SAS, SPSS) for implementation of multivariate analysis methods.
6. Select appropriate multivariate methods for use in common multivariate problems encountered in applications.
7. Prepare a written and oral report and communicate the findings to a technical and non-technical audience.

Examination and Assessment

Total Marks 200

- Continuous Assessment 200 Marks
 - Assignment - Detailed breakdown will be provided at beginning of the module (200 Marks)

Penalties (for late submission of Course/Project Work etc.)

Work which is submitted late shall be assigned a mark of zero (or a Fail Judgement in the case of Pass/Fail modules)

Pass Standard

40%

Special Requirements for Passing Module

None

Supplemental Examination and Assessment

- Continuous Assessment
 - Marks in passed elements of continuous assessment are carried forward. Failed elements must be repeated
 - To be prescribed by module coordinator